

California Housing: Linear Regression

Short model report

Objective

Predict median owner-occupied house value using eight California block-group features from scikit-learn's dataset.

Method

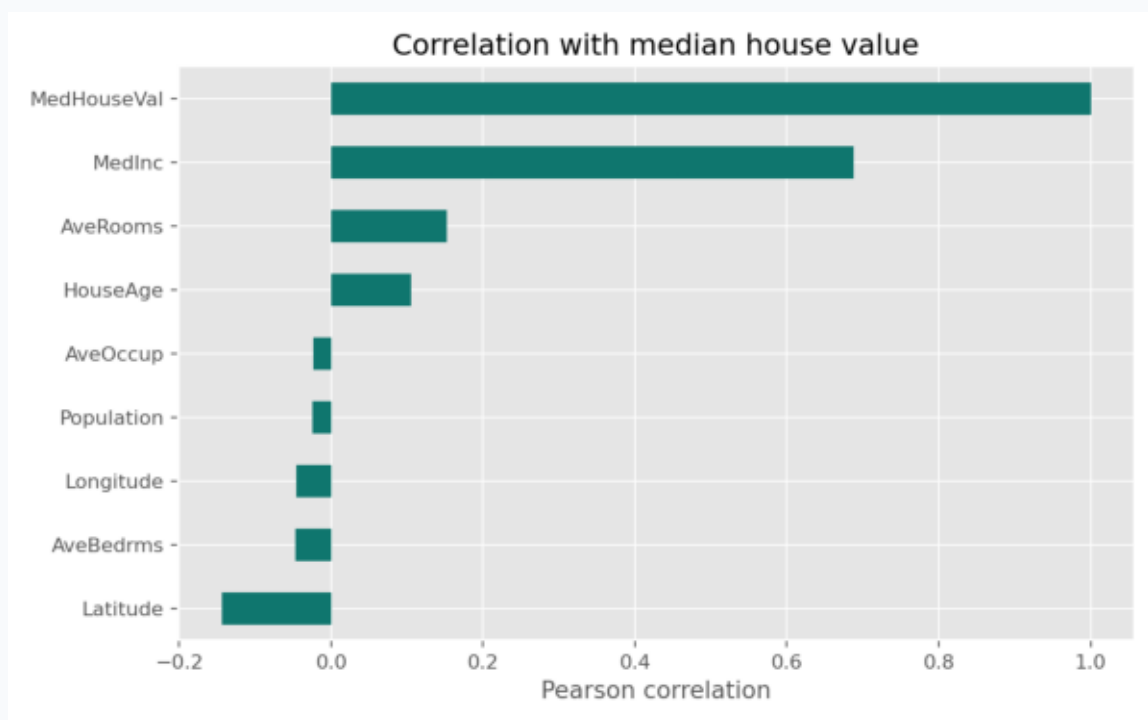
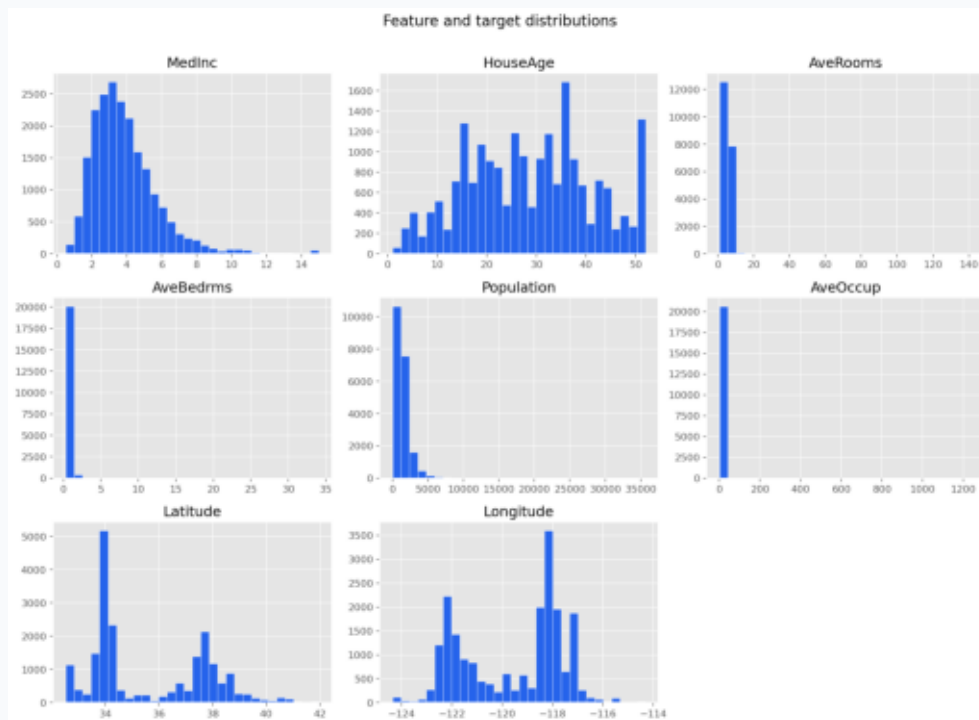
- 20,640 rows; 80/20 train-test split (seed 42)
- Invalid/blank numeric values coerced to missing
- Median imputation + LinearRegression in one pipeline
- Evaluation performed on 4,128 held-out rows

MAE 0.5332 100k **RMSE 0.7456 100k** **R² 0.5758**

Target values are in \$100,000s of 1990 USD. This is an educational baseline, not a valuation tool.

Exploratory data analysis

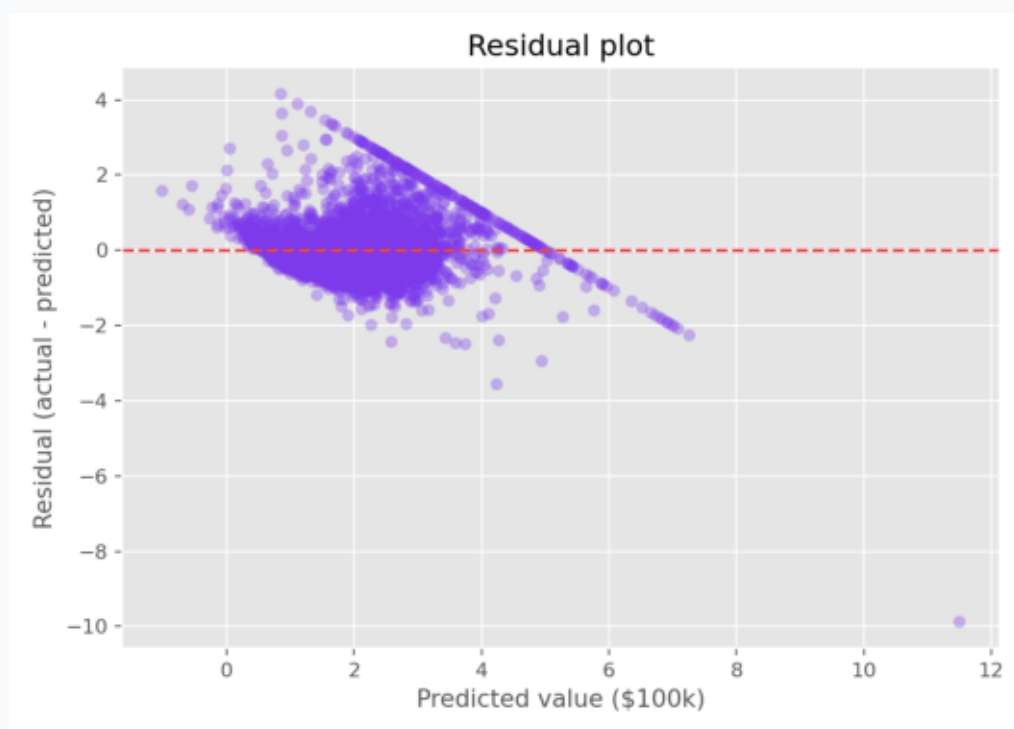
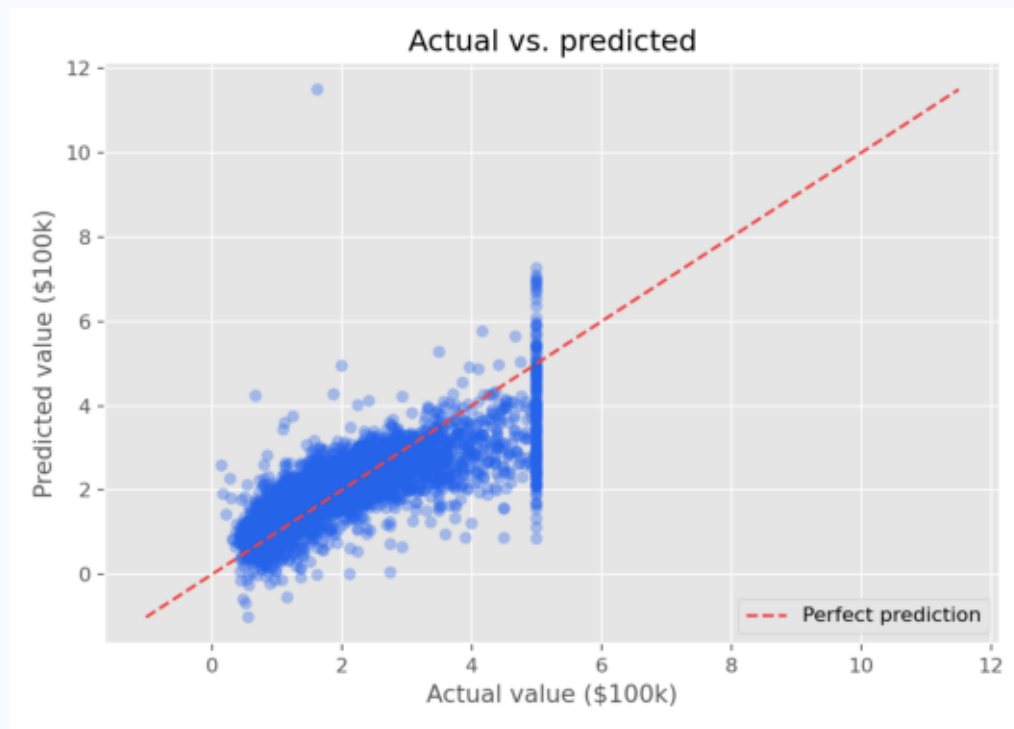
Distributions and feature relationships



Median income and location-related features show useful association with the target; the target's cap makes

Evaluation and improvement ideas

Held-out predictions and residual diagnostics



Next: compare Ridge/Lasso and tree ensembles with cross-validation; use newer, property-level data and ass